

CSCE 647 IMAGE SYNTHESIS · MODULE 09 · LECTURE NOTES

Lights and Sampling Them

Where the energy comes from, and how to find it.

MODULE OUTCOMES

Implement area, point, directional, and environment lights with correct PDFs.

Sample a sphere light by solid angle rather than by area.

Build a 2D CDF for environment map importance sampling.

Explain the many-lights problem and the structures that solve it.

Explain why light sampling PDFs must be expressed in a consistent measure.

1 Kinds of light

Type	Definition	Physical?	Sampling
Point	All energy emitted from a single position.	No. Infinite radiance from zero area.	One direction, probability 1.
Spot	A point light with an angular falloff.	No, for the same reason.	As point.
Directional	Parallel rays from infinitely far away.	An idealisation of a very distant source.	One direction, probability 1.
Area	Emitting geometry with finite extent.	Yes. This is what real lights are.	Sample a point on the surface.
Environment	Radiance arriving from every direction at infinity; an image.	An idealisation of distant surroundings.	2D CDF over luminance (§3).

Point and directional lights are conveniences inherited from an era when area lights were unaffordable. They produce perfectly sharp shadow boundaries, which no real light source does, and they are the main reason early computer graphics has its characteristic look. Support them — they are useful and artists expect them — but understand that an area light is the physical object.

DELTA LIGHTS ARE A SPECIAL CASE THROUGH THE WHOLE INTEGRATOR

They cannot be hit by a BSDF-sampled ray. The probability of a randomly chosen direction passing exactly through a point is zero, so a delta light exists in the image *only* through next-event estimation.

Their MIS weight is therefore always exactly 1. The BSDF strategy could never have generated that direction, so its share of the density is zero. This is the mirror image of the specular-BSDF case in Module 08: there the light strategy had zero density; here the BSDF strategy does.

They cannot be seen directly by a camera ray, so they need no geometry and contribute nothing when looked at.

Flag them explicitly, exactly as Module 07 flagged delta BSDFs. Every place the integrator asks 'could the other strategy have produced this sample?' requires the answer, and deriving it from the light type at each site is how inconsistencies creep in.

2 Sampling area lights

An area light is sampled by choosing a point on its surface. But the integrator works in solid angle, so the density must be converted.

$$p(\omega) = p(A) \cdot |p - q|^2 / |\cos\theta_q|$$

This is the geometry term from Module 02 appearing in a new role: the Jacobian relating a density over the light's area to a density over directions at the shading point.

USE ONE MEASURE THROUGHOUT, AND CONVERT AT THE BOUNDARY

This is the hardest class of bug in light sampling, because it produces a plausible image that is systematically wrong by a factor varying with distance and orientation.

An MIS weight combines two densities, and they must be in the same measure. If the light sampler reports a density per unit area and the BSDF reports one per unit solid angle, the weight is meaningless and the result is biased — but it converges smoothly to the wrong answer, so it looks like success.

Pick solid angle for the entire renderer. Every pdf() returns a solid-angle density; light samplers convert internally before returning. Then the question can never arise at a call site.

The test: render a scene with a light at distance 1 and the same scene scaled up by 100. The converged images must be identical. If they differ, a measure conversion is missing.

SAMPLE A SPHERE LIGHT BY SOLID ANGLE, NOT BY AREA

The obvious implementation samples uniformly over the sphere's surface. **Half the samples land on the far side**, facing away from the shading point, and contribute exactly nothing.

It is worse than half. Samples near the visible silhouette have a grazing $\cos\theta_q$ and contribute almost nothing either, so the genuinely useful fraction is well under 50%.

Sample the cone of directions the sphere subtends instead. Every direction in that cone strikes the visible hemisphere, and the density is uniform over the cone, which is a closed-form construction of about ten lines.

Typically two to four times less variance, with shorter code. The general principle — sample the directions that can contribute, rather than the geometry that may not — applies to every closed light shape, and Ureña's solid-angle quad sampling does the same for rectangles.

Shape	Method	Watch for
Triangle	Uniform barycentric sampling using $\sqrt{u_1}$.	Reject when the shading point is behind the light's plane — otherwise you sample a light that cannot illuminate it.
Quad	Uniform in 2D, or solid-angle sampling (Ureña et al. 2013).	Solid-angle sampling is a large win for big nearby quads — softboxes, windows — and roughly neutral for small distant ones.
Sphere	Cone sampling, as above.	Must degenerate correctly to full-sphere sampling when the shading point is inside the sphere.
Disk	$\sqrt{}$ for the radius, uniform in angle.	The same $\sqrt{}$ as Module 06.
Mesh light	Select a triangle with probability proportional to area $\times$ emitted radiance, then sample within it.	Build the selection CDF once at load, not per sample.

3 Environment lights

An environment light surrounds the scene with an image — typically a high-dynamic-range photograph in equirectangular projection — providing radiance from every direction. It is the standard way to light a scene realistically, and it is useless without importance sampling.

WHY UNIFORM SAMPLING OF AN ENVIRONMENT MAP FAILS COMPLETELY

A typical outdoor HDRI has the sun occupying on the order of 0.001% of its pixels while carrying the **majority of the total energy** — a dynamic range of 10^5 or more between the sun and the sky beside it.

Uniform directional sampling finds the sun roughly once in 100,000 samples, and returns an enormous value when it does.

That is a firefly generator, not a light source. The image is unusable at any practical sample count, and no amount of waiting fixes it, because the variance is dominated by an event that almost never occurs.

With a luminance-weighted CDF, the sun is sampled roughly in proportion to the energy it carries. **This single data structure is the difference between HDRI lighting working and not working**, and it is about eighty lines.

```
// Build once at load.
for (int y = 0; y < H; y++) {
    float sinTheta = sinf(PI * (y + 0.5f) / H);    // area distortion
    for (int x = 0; x < W; x++)
        f[y][x] = luminance(env(x,y)) * sinTheta;
    rowCdf[y] = build_1d_cdf(f[y]);                // conditional p(x|y)
}
marginalCdf = build_1d_cdf(rowSums);              // marginal p(y)

// Sample: marginal then conditional. Two binary searches.
auto [y, pdf_y] = sample_1d(marginalCdf, u1);
auto [x, pdf_x] = sample_1d(rowCdf[y], u2);
float sinT = sinf(PI * (y + 0.5f) / H);
float pdf = (pdf_x * pdf_y * W * H) / (2 * PI * PI * sinT);
```

THE TWO $\sin\theta$ FACTORS MEAN DIFFERENT THINGS

In the build: an equirectangular map distorts area. A pixel row near the pole covers far less solid angle than one at the equator, despite having the same number of pixels. Weighting by $\sin\theta$ corrects for this, so the CDF is proportional to *energy* rather than to pixel value.

In the PDF: the sampling procedure produced a density over image coordinates, and the integrator needs one over solid angle. Dividing by $\sin\theta$ performs that measure conversion — §2's rule again.

They are separate corrections that happen to involve the same function, and conflating them — or applying one and not the other — is the usual bug here. The symptom is an environment light that is systematically wrong near the poles, which in an outdoor scene means the sky directly overhead.

The test: render a diffuse white sphere under a constant white environment and compare against the analytic answer. It must match exactly, because a constant environment has an analytic solution.

4 Many lights

THE PROBLEM

Next-event estimation needs a light to sample. With N lights, selecting one uniformly and dividing by $1/N$ is unbiased, and costs roughly a factor of N in variance — the one light that actually illuminates the shading point is chosen only $1/N$ of the time.

A city at night, an interior with hundreds of fixtures, or a scene with emissive geometry can have 10^4 to 10^6 emitters. **Most contribute nothing at all to any given shading point:** they are behind a wall, facing away, or far enough that their contribution is negligible.

Sampling all of them is prohibitively expensive. Sampling one uniformly is prohibitively noisy.

The answer is a structure that selects lights approximately in proportion to their actual contribution *at the shading point being considered* — which is a different distribution at every point.

Method	Idea	Assessment
Power sampling	Build one global CDF over total emitted power.	Trivial, and better than uniform. Ignores distance and visibility entirely, so a powerful light on the other side of the building is still chosen often.
Light BVH	A hierarchy over the lights, each node storing bounds, total power, and an orientation cone. Descend stochastically, choosing children in proportion to an estimate of their contribution at the shading point.	The standard answer in production renderers. Logarithmic in light count, adapts per shading point, and the selection probability is exactly computable — which §4's constraint requires.
Lightcuts	Cluster lights into a hierarchy and compute an error bound for treating a whole cluster as one representative light. Refine until the bound is acceptable.	Gives a genuine error bound rather than a heuristic. More complex, and the bounds can be loose.
ReSTIR	Resampled importance sampling with reservoirs, reusing samples across neighbouring pixels and across frames.	Transformative. Makes millions of lights tractable in real time. Arguably the most significant rendering result of the last decade. Module 12 returns to it.

WHATEVER THE SCHEME, THE PROBABILITY MUST BE EXACT

Every selection scheme must report **the exact probability with which it chose the light it chose**. That probability divides the contribution in the estimator and feeds the MIS weight in Module 08.

If the reported probability does not match the actual selection behaviour, the result is biased — and, as always, biased in a way that converges confidently. A light BVH whose descent probabilities do not match its stated PDF produces a smooth, wrong image.

A light with zero selection probability is a light that does not exist. Any heuristic that prunes 'unimportant' lights outright introduces bias. Schemes must give every potentially contributing light a non-zero chance, however small.

This is Module 06's defensive sampling rule applied to light selection, and it is why light hierarchies use *stochastic* traversal rather than descending into the apparently best child.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
PBRT 4th ed. — Chapter 12, Light Sources	Every light type with its sampling routine and PDF, including the environment map 2D distribution of §3.
PBRT 4th ed. — §12.6, Light Sampling (BVH)	The light BVH of §4, with the importance heuristic and the orientation cone bounds.
Ureña, Fajardo & King — An Area-Preserving Parametrization for Spherical Rectangles (free)	Solid-angle sampling for quads. Arnold's research page has a great deal of production-grade free material.
Bitterli et al. — Spatiotemporal Reservoir Resampling (ReSTIR, free)	The many-lights result of §4. Read it after Module 12.
Poly Haven — free HDRIs (CC0)	High-quality environment maps, CC0 licensed, for testing §3.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement area lights with triangle and quad sampling. Verify the PDF with a χ^2 test in solid-angle measure.
- 2 Implement sphere light sampling by area and by subtended cone. Report the variance ratio at several distances.

- 3 Perform the scale test from §2: render a scene, then the same scene scaled by 100×, and confirm the converged images match. If they do not, find the missing measure conversion.
- 4 Build the 2D CDF for an environment map. Visualise the sampling density as an image and confirm the sun is bright in it.
- 5 Render a scene with an outdoor HDRI with and without environment importance sampling at 64 samples per pixel. The difference is the entire point of §3.
- 6 Verify the constant-environment test: a diffuse white sphere under a uniform environment must match the analytic answer exactly.
- 7 Implement power-proportional light selection and compare against uniform selection on a scene with 100 lights of widely varying brightness.
- 8 Implement a simple light BVH and compare against power sampling on a scene with 10,000 lights spread over a large area. Report variance and time.
- 9 Deliberately prune lights below a contribution threshold and demonstrate the resulting bias with a convergence plot.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the five light types and say which are physical.
- 2 Give three ways delta lights are special in the integrator.
- 3 Write the conversion between area-measure and solid-angle-measure densities.
- 4 Why must the whole renderer use one measure, and what test detects a mixed-measure bug?
- 5 Why is cone sampling better than area sampling for a sphere light?
- 6 Why does uniform sampling of an environment map fail, and what does the 2D CDF fix?
- 7 Explain the two different roles $\sin\theta$ plays in environment map sampling.
- 8 What is the many-lights problem, and why does uniform selection cost a factor of N ?
- 9 What must every light selection scheme report, and why is pruning a light biased?